

Big Data e Computação em Nuvem

Spark ML

The logo for Insper, featuring the word "Insper" in a black serif font. The text is centered within a white rectangular area that has a red border on its top and left sides. The background of the slide features a vertical rainbow gradient on the right side.

Insper

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Panorama da disciplina

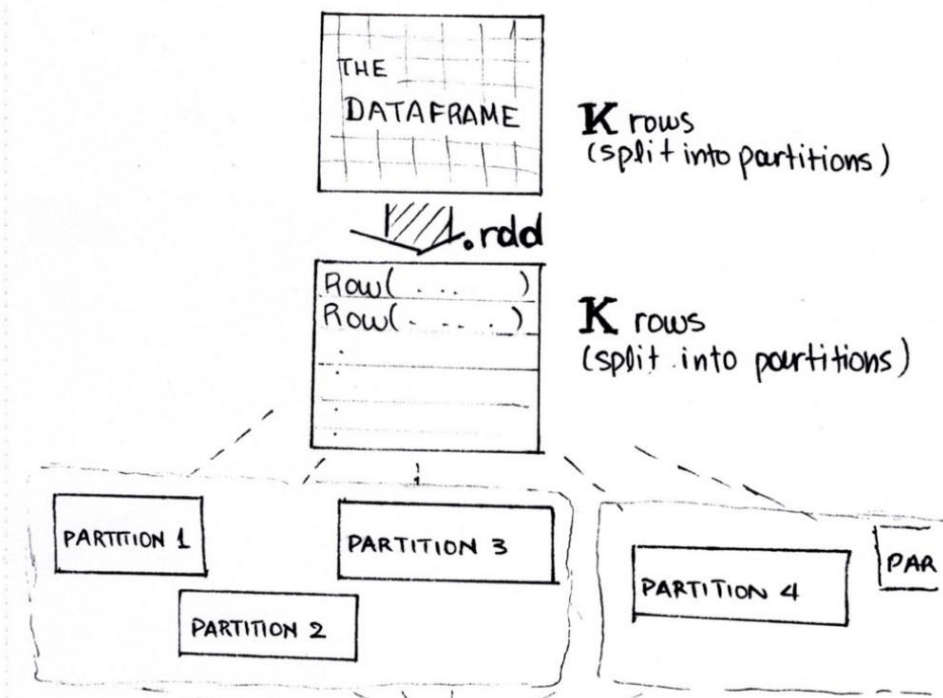
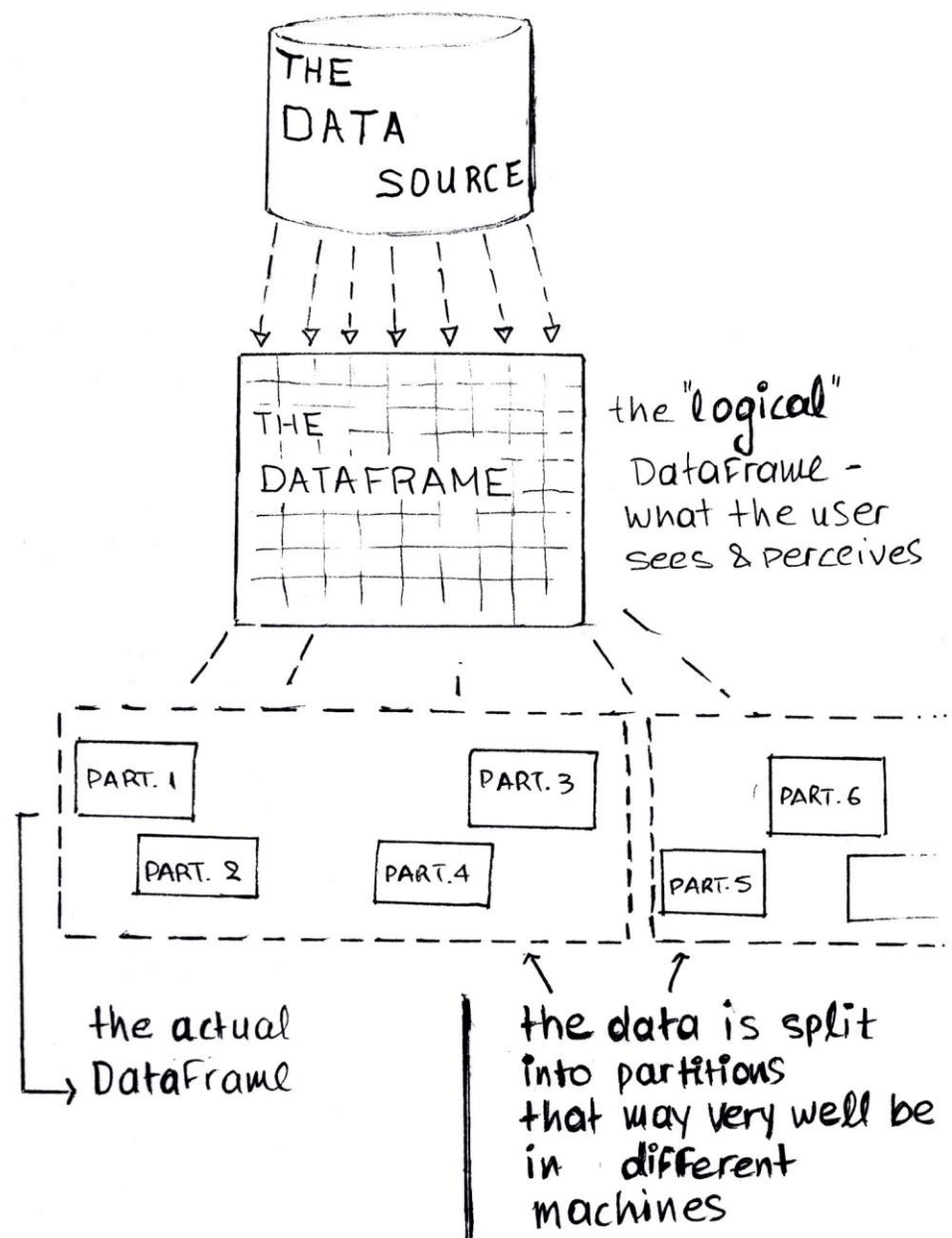
O que veremos?

- **Aula 8 [10/nov]:**
 - DataFrames em análise descritiva de datasets reais
 - Projeto final
- **Aula 9 [17/nov]:**
 - DataFrame Otimizado
 - Spark Machine Learning
 - Spark Pipelines
 - Projeto final
- **Aula 10 [24/nov]:**
 - ML em datasets reais
 - Projeto final
- **Aula 11 [28/nov]:**
 - Cloud
 - Databricks
 - Projeto final
- **Aula 12 [29/nov]:**
 - Checkpoint
- **Aula 13 [01/dez]:**
 - Projeto final
- **Aula 14 [08/dez]:**
 - Apresentação do projeto final

DataFrame avançado

Otimizando o uso de dados em Spark

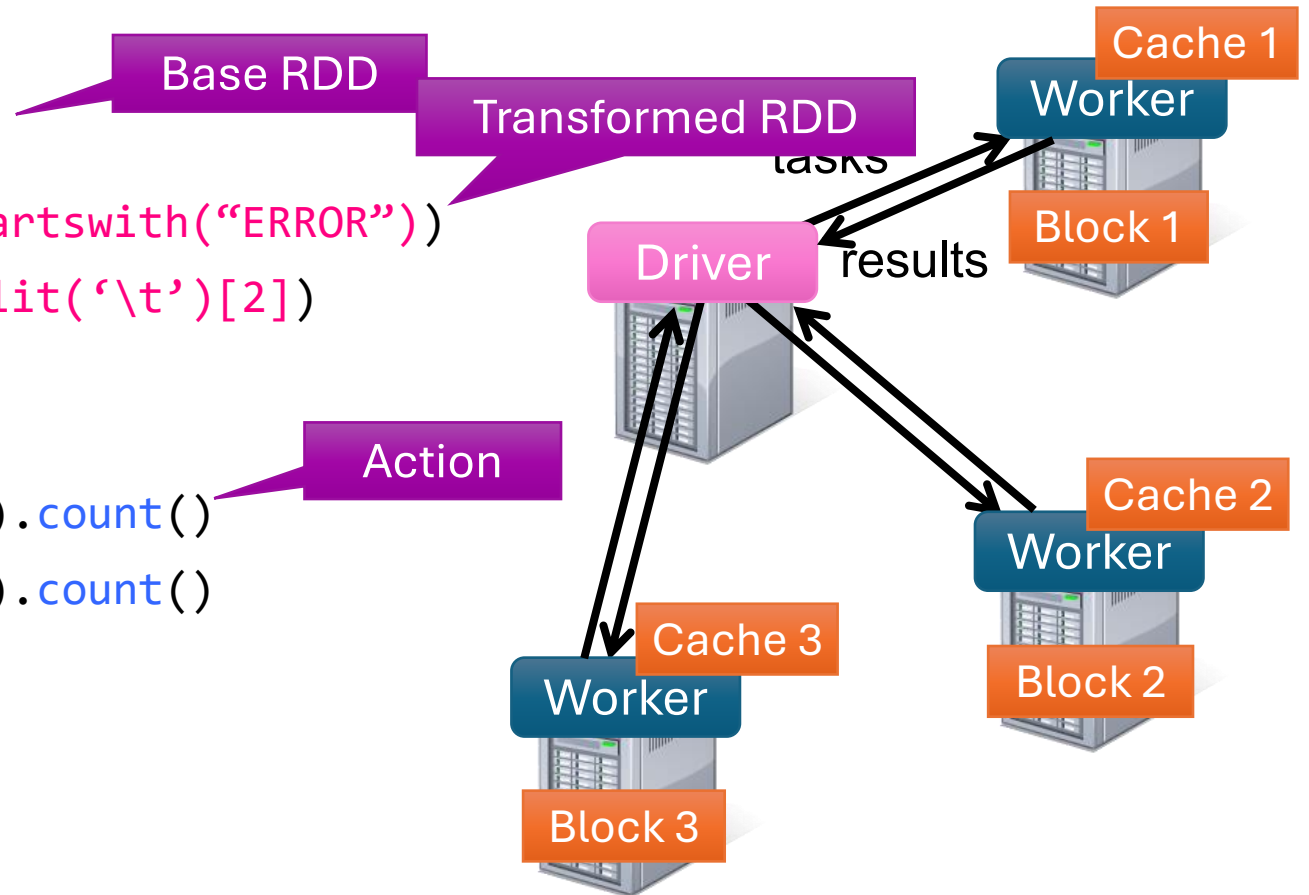
DataFrame



Spark – Mecanismo de Cache

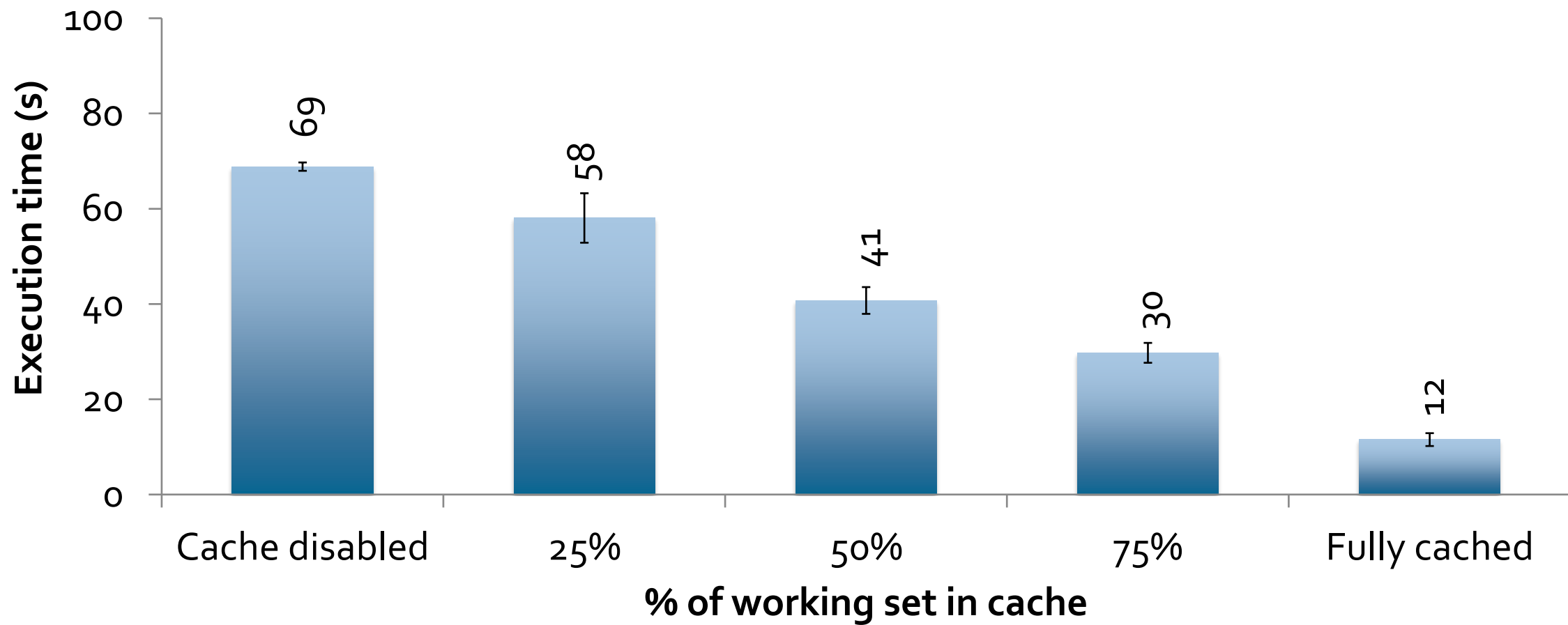
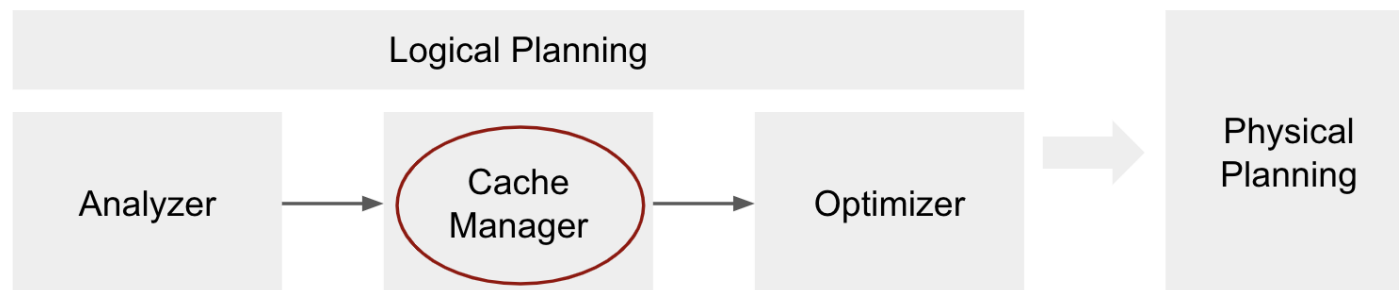
```
lines = spark.textFile("hdfs://...")
errors = lines.filter(lambda s: s.startswith("ERROR"))
messages = errors.map(lambda s: s.split('\t')[2])
messages.cache()

messages.filter(lambda s: "foo" in s).count()
messages.filter(lambda s: "bar" in s).count()
. . .
```



Scaling Down

Uso de Cache



Uso do cache

Para pensar

```
inputRDD = sc.textFile("words.txt").cache()

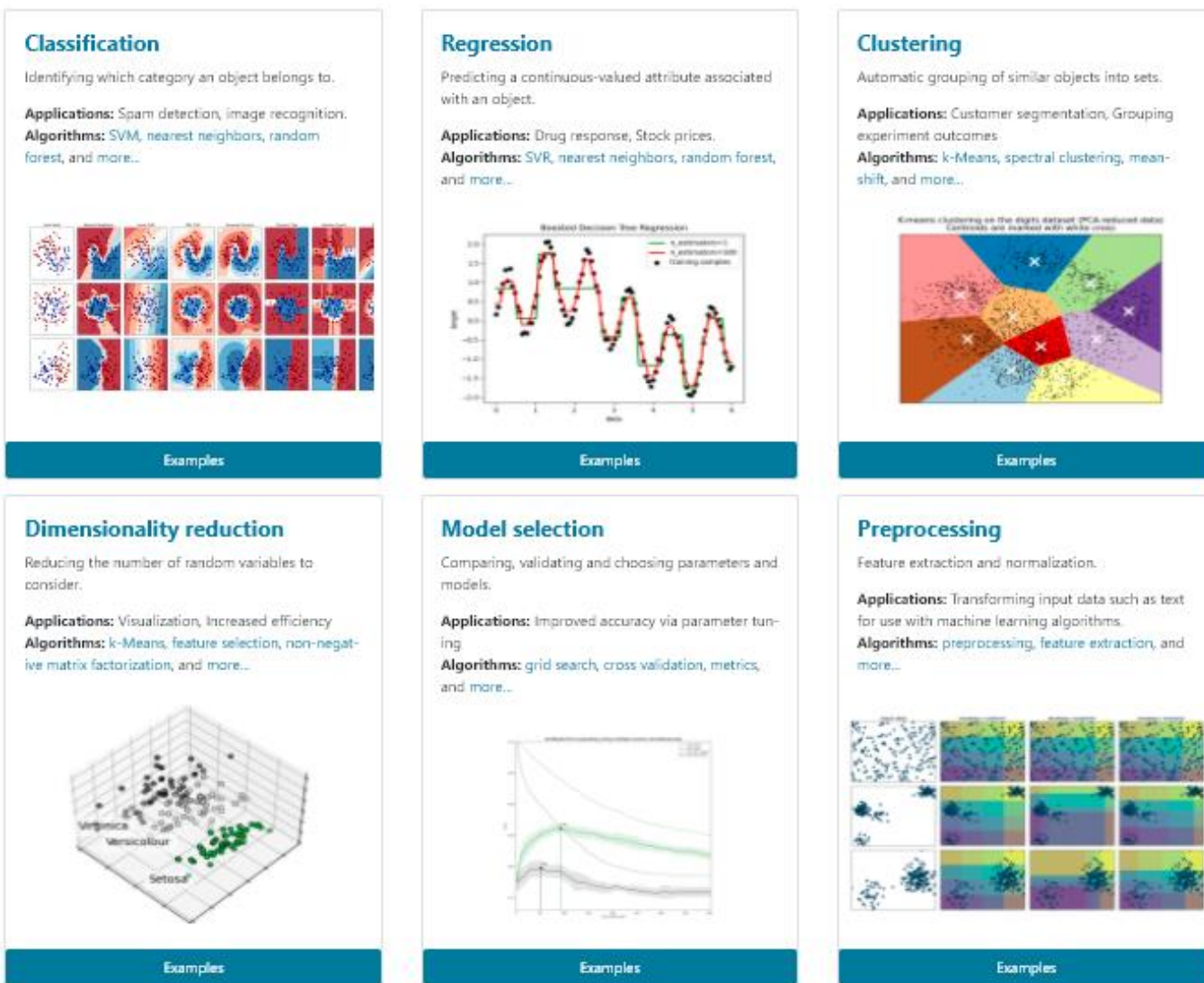
print("Number of words: ",inputRDD.count())
print("Number of distinct words: ", inputRDD.distinct().count())
```

Spark ML

Fazendo previsões com Spark

scikit-learn (sklearn)

Principal biblioteca Python para ML



News

On-going development: [What's new \(Changelog\)](#)

October 2021. scikit-learn 1.0.1 is available for download ([Changelog](#)).

September 2021. scikit-learn 1.0 is available for download ([Changelog](#)).

April 2021. scikit-learn 0.24.2 is available for download ([Changelog](#)).

January 2021. scikit-learn 0.24.1 is available for download ([Changelog](#)).

December 2020. scikit-learn 0.24.0 is available for download ([Changelog](#)).

August 2020. scikit-learn 0.23.2 is available for download ([Changelog](#)).

May 2020. scikit-learn 0.23.1 is available for download ([Changelog](#)).

May 2020. scikit-learn 0.23.0 is available for download ([Changelog](#)).

Scikit-learn from 0.23 requires Python 3.6 or newer.

March 2020. scikit-learn 0.22.2 is available for download ([Changelog](#)).

January 2020. scikit-learn 0.22.1 is available for download ([Changelog](#)).

December 2019. scikit-learn 0.22 is available for download ([Changelog](#) and [Release Highlights](#)).

scikit-learn (sklearn)

Principal biblioteca Python para ML

```
import matplotlib.pyplot as plt
import numpy as np
from sklearn import datasets, linear_model
from sklearn.metrics import mean_squared_error, r2_score

# Load the diabetes dataset and use only one feature
diabetes_X, diabetes_y = datasets.load_diabetes(return_X_y=True)
diabetes_X = diabetes_X[:, np.newaxis, 2]

# Split the DATA and TARGETS into training/testing sets
diabetes_X_train = diabetes_X[:-20]
diabetes_X_test = diabetes_X[-20:]
diabetes_y_train = diabetes_y[:-20]
diabetes_y_test = diabetes_y[-20:]

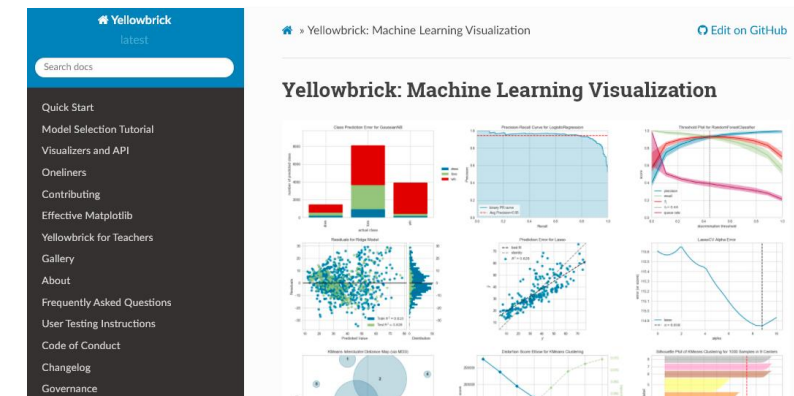
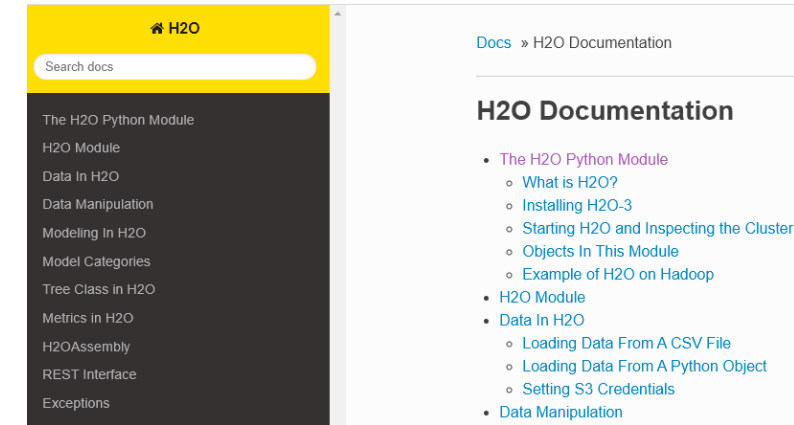
# Create linear regression object
regr = linear_model.LinearRegression()

# Train the model using the training sets
regr.fit(diabetes_X_train, diabetes_y_train)

# Make predictions using the testing set
diabetes_y_pred = regr.predict(diabetes_X_test)

# The coefficients
print("Coefficients: \n", regr.coef_)
# The mean squared error
print("Mean squared error: %.2f" % mean_squared_error(diabetes_y_test, diabetes_y_pred))
# The coefficient of determination: 1 is perfect prediction
print("Coefficient of determination: %.2f" % r2_score(diabetes_y_test, diabetes_y_pred))
```

Code source: Jaques Grobler | License: BSD 3 clause



SparkML

Machine Learning em Spark

[Overview](#) [Programming Guides ▾](#) [API Docs ▾](#) [Deploying ▾](#) [More ▾](#)

spark.ml package

- [Overview: estimators, transformers and pipelines](#)
- [Extracting, transforming and selecting features](#)
- [Classification and Regression](#)
- [Clustering](#)
- [Advanced topics](#)

spark.mllib package

- [Data types](#)
- [Basic statistics](#)
- [Classification and regression](#)
- [Collaborative filtering](#)
- [Clustering](#)
- [Dimensionality reduction](#)
- [Feature extraction and transformation](#)
- [Frequent pattern mining](#)
- [Evaluation metrics](#)

Overview: estimators, transformers and pipelines - spark.ml

The `spark.ml` package aims to provide a uniform set of high-level APIs built on top of [DataFrames](#) that help users create and tune practical machine learning pipelines. See the [algorithm guides](#) section below for guides on sub-packages of `spark.ml`, including feature transformers unique to the Pipelines API, ensembles, and more.

Table of contents

- [Main concepts in Pipelines](#)
 - [DataFrame](#)
 - [Pipeline components](#)
 - [Transformers](#)
 - [Estimators](#)
 - [Properties of pipeline components](#)
 - [Pipeline](#)
 - [How it works](#)
 - [Details](#)
 - [Parameters](#)
 - [Saving and Loading Pipelines](#)
- [Code examples](#)
 - [Example: Estimator, Transformer, and Param](#)
 - [Example: Pipeline](#)
 - [Example: model selection via cross-validation](#)
 - [Example: model selection via train validation split](#)

SparkML

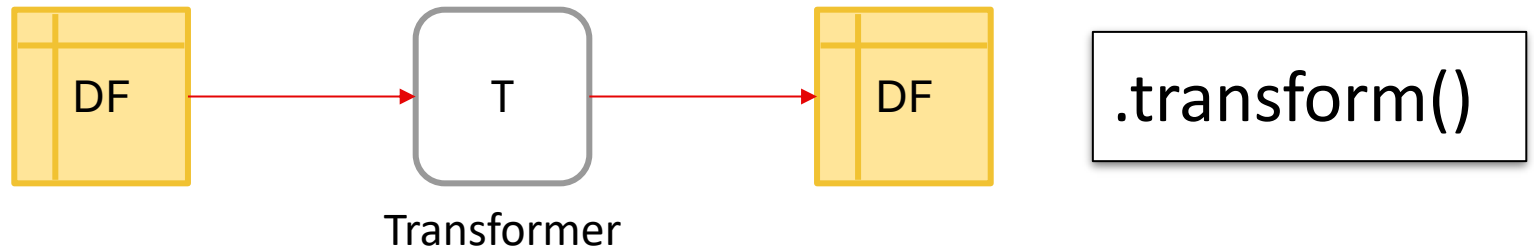
Principais componentes

- DataFrame
- Transformer
- Estimator
- Pipeline
- Parameter

SparkML

Principais componentes

- DataFrame
- **Transformer**
- Estimator
- Pipeline
- Parameter



This section covers algorithms for working with features, roughly divided into these groups:

- Extraction: Extracting features from "raw" data
- Transformation: Scaling, converting, or modifying features
- Selection: Selecting a subset from a larger set of features

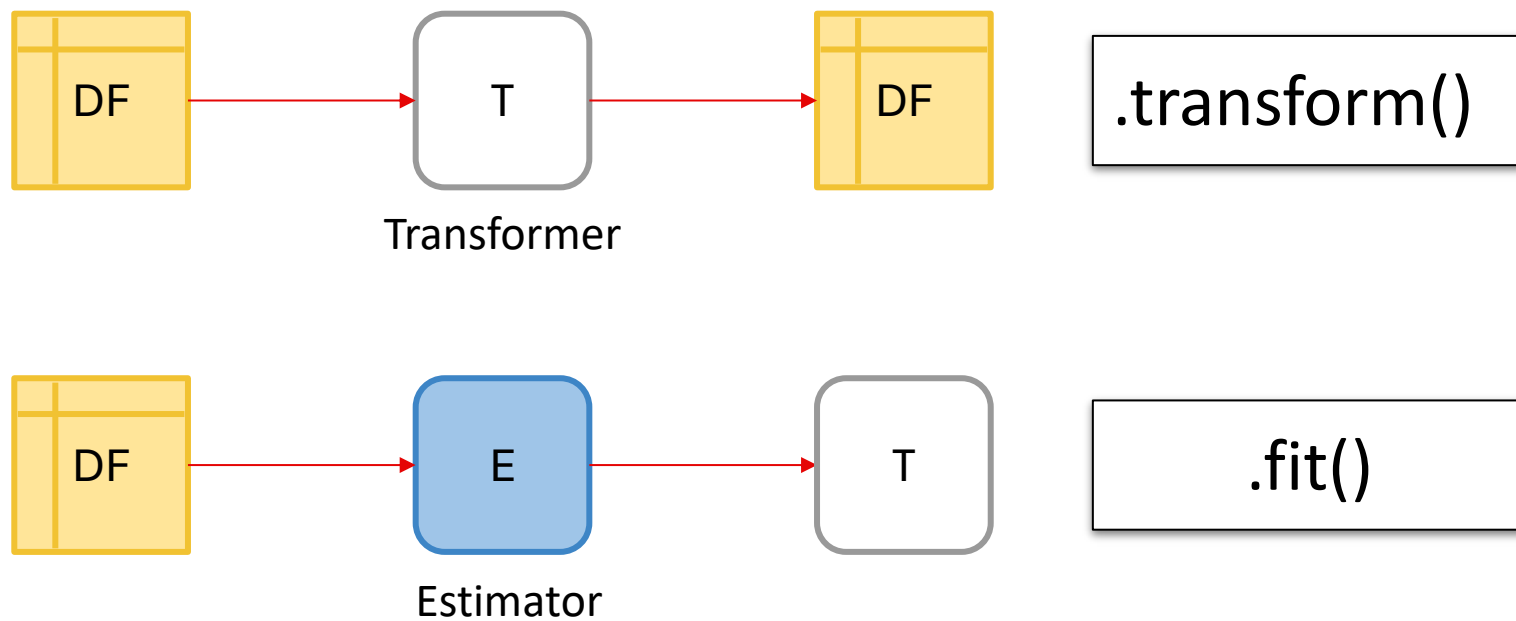
Table of Contents

- Feature Extractors
 - TF-IDF (HashingTF and IDF)
 - Word2Vec
- Feature Transformers
 - Tokenizer
 - Binarizer
 - PolynomialExpansion
 - StringIndexer
 - OneHotEncoder
 - VectorIndexer
 - Normalizer
 - StandardScaler
 - Bucketizer
 - ElementwiseProduct
 - VectorAssembler
- Feature Selectors

SparkML

Principais componentes

- DataFrame
- Transformer
- **Estimator**
- Pipeline
- Parameter



SparkML

Principaux composants

- DataFrame
- Transformer
- **Estimator**
- Pipeline
- Parameter

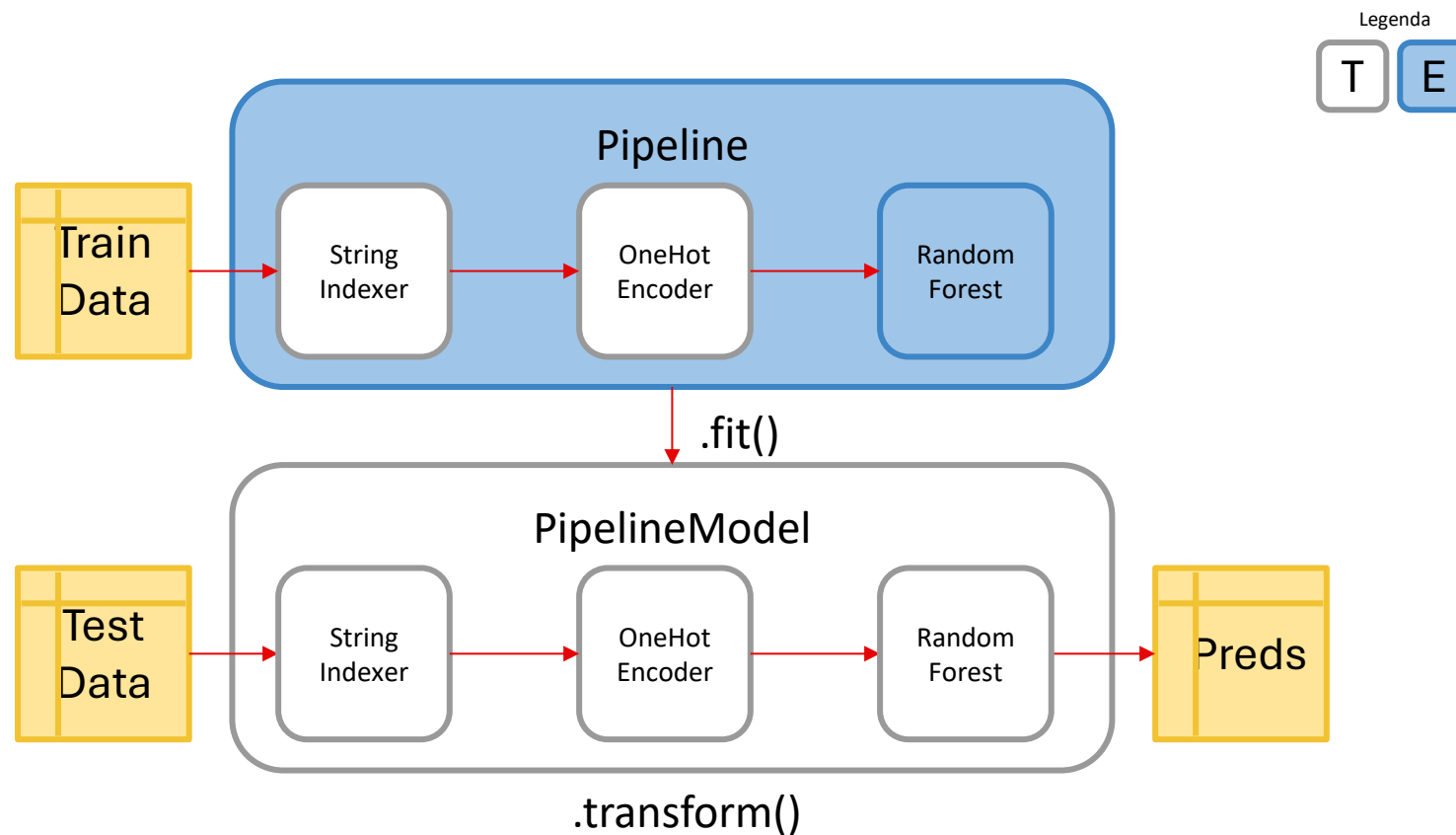
Table of Contents

- Classification
 - Logistic regression
 - Binomial logistic regression
 - Multinomial logistic regression
 - Decision tree classifier
 - Random forest classifier
 - Gradient-boosted tree classifier
 - Multilayer perceptron classifier
 - Linear Support Vector Machine
 - One-vs-Rest classifier (a.k.a. One-vs-All)
 - Naïve Bayes
 - Factorization machines classifier
- Regression
 - Linear regression
 - Generalized linear regression
 - Available families
 - Decision tree regression
 - Random forest regression
 - Gradient-boosted tree regression
 - Survival regression
 - Isotonic regression
 - Factorization machines regressor
- Linear methods
- Factorization Machines
- Decision trees
 - Inputs and Outputs
 - Input Columns
 - Output Columns
- Tree Ensembles
 - Random Forests
 - Inputs and Outputs
 - Input Columns
 - Output Columns (Predictions)
 - Gradient-Boosted Trees (GBTs)
 - Inputs and Outputs
 - Input Columns
 - Output Columns (Predictions)

SparkML

Principais componentes | Pipeline durante o desenvolvimento

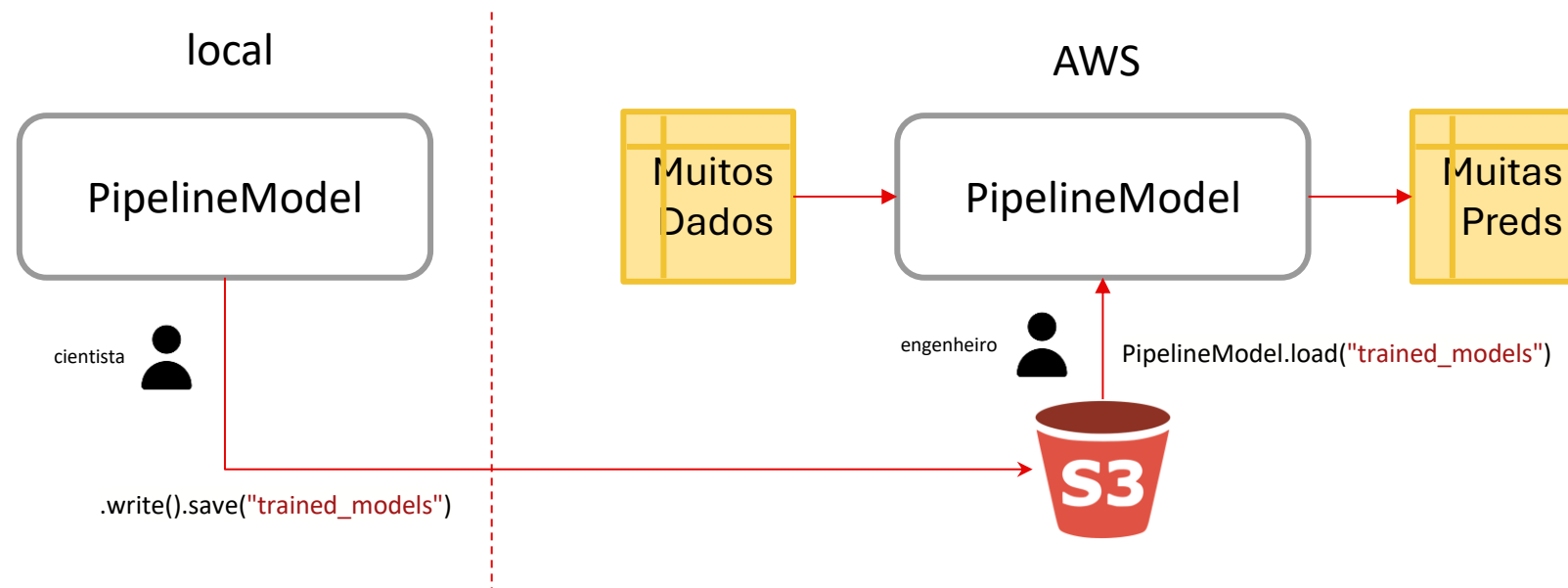
- DataFrame
- Transformer
- Estimator
- **Pipeline**
- Parameter



SparkML

Principais componentes | Pipeline em produção

- DataFrame
- Transformer
- Estimator
- **Pipeline**
- Parameter



SparkML

Links para documentações relevantes

- <https://spark.apache.org/docs/latest/ml-guide.html>
- <https://spark.apache.org/docs/latest/ml-features>
- <https://spark.apache.org/docs/latest/ml-classification-regression.html>



Prática!

Criando modelos preditivos no Spark

- Modelo de regressão linear
- Usando Pipeline
- Feature engineering
- Salvando/Carregando modelos
- Otimização de hiperparâmetros e Validação cruzada

Obrigado!

